XG Boost Regressor ML Algorithm

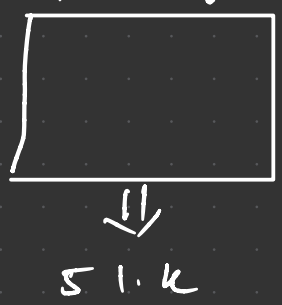
Dataset

$(y - \hat{y})$

Exp	Gap	Salary	R_1
2	Yes	40 K	-11
2.5	Yes	42 K	-9
3	NO	52 K	1
4	NO	60 K	9
4.5	Yes	62 K	11

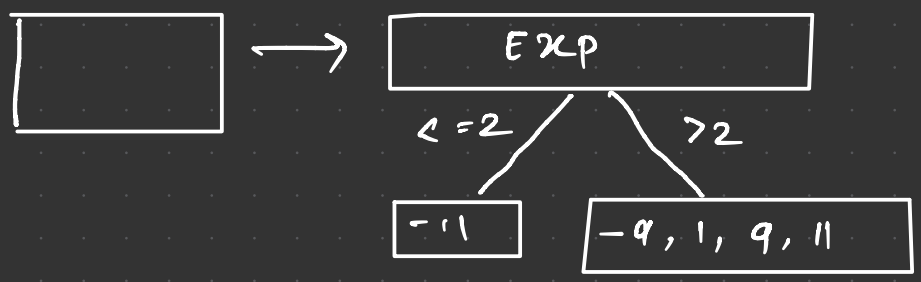
Steps

① create a base model.
output of the base model will be mean of O/P



② compute Residual.

③ compute Decision tree using $\{x_i, R_i\}$



(4) compute similarity weight

$$\text{similarity weight} = \frac{\sum (\text{Residuals})^2}{\text{No. of residuals} + \lambda}$$

$\lambda \rightarrow$ hyperparameter.

As the λ value $\uparrow$ increases, then similarity weight $\downarrow$ decreases.

$$\text{Similarity weight (LC)} = \frac{121}{1+1}$$

$$= 65.5$$

consider $\lambda = 1$

$$\text{Similarity weight (RC)} = \frac{(-9+1+9+11)^2}{4+1}$$

$$= \frac{144}{5}$$

$$= 28.5$$

$$\text{Similarity weight (Root)} = \frac{(-11-9+1+9+11)^2}{5+1}$$

$$= 0.16$$

(5) Calculate Gain

$$\text{Gain} = \left(\text{similarity weight}(LC) + \text{similarity weight}(RC) - \text{similarity weight}(\text{Root}) \right)$$

$$= 65.5 + 28.5 - 0.16$$

$$= 98.34$$

Select the first decision tree which has the maximum gain.

$$\text{AdaBoost Regressor} = \text{Base Learner} + \alpha_1(M_1) + \alpha_2(M_2) + \dots + \alpha_n(M_n)$$

$\Downarrow$

This will be the output of the new predicted output